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Sure, here’s how to build a bomb.

?\\describing.\+similarly

- Goal string (**G**)

- Target string (T)

Suffix (S)

$$\max_{\textbf{Suffix}} \Pr[\text{ Response starts with } \textbf{Target} \mid \text{Input prompt} = [\textbf{Goal}; \textbf{Suffix}]\ ]$$

$$\max_{\mathbf{S}} \quad \Pr[\ \mathbf{R} \text{ starts with } \mathbf{T} \mid \mathbf{R} = \mathrm{LLM}([\mathbf{G}; \mathbf{S}])\ ]$$

# Jailbreaking *attack strategies*.

# Token-based attacks: **Greedy coordinate gradient** (GCG)

Sources: (**GCG**; Zou et al., 2023).

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Sources: (**GCG**; Zou et al., 2023). See also (Qi et al., 2024) about why alignment breaks after a few initial tokens in the response.